

AZ-900

Learning path 4: Apply Azure Skills in Guided Projects



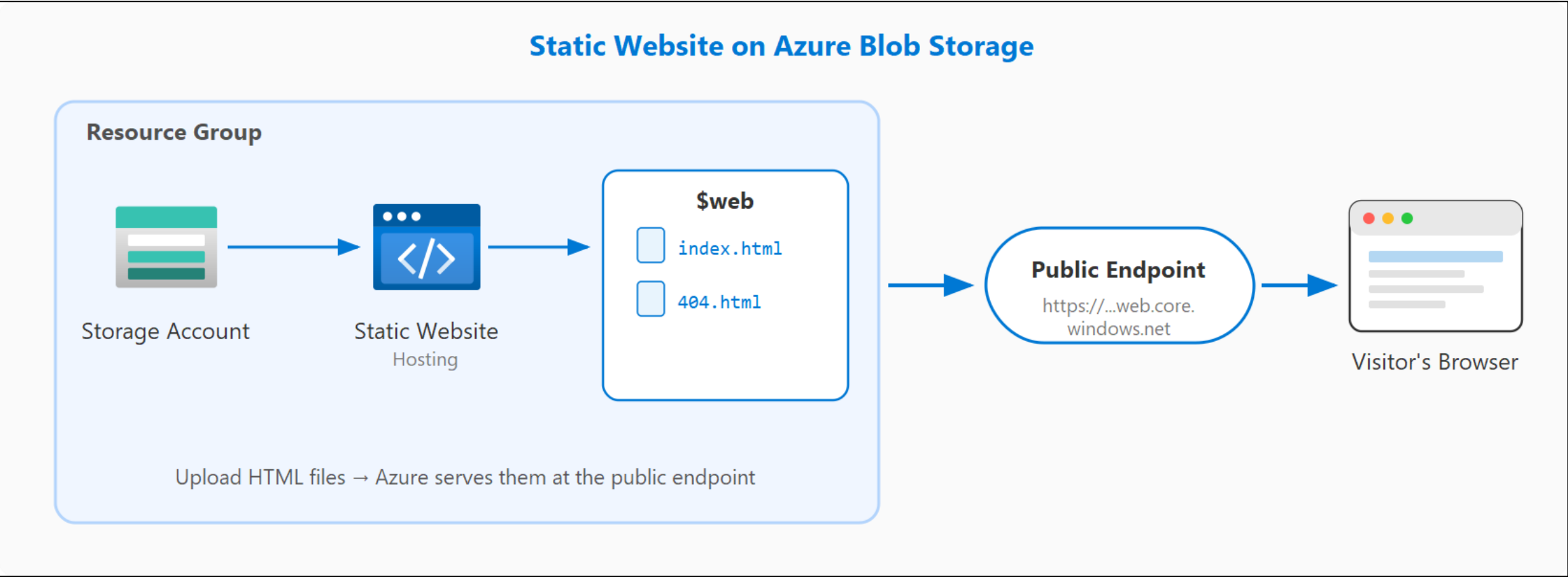
Guided project: Deploy a static website with Azure Blob Storage



Project: Static website on Blob Storage objectives

- Scenario: Publish a product landing page without provisioning VMs or web servers
- Enable static website hosting on a storage account to create a public endpoint
- Azure automatically creates a \$web container to serve HTML, CSS, and image files
- Exercise 1: Create a storage account and enable static website hosting
- Exercise 2: Upload HTML content and verify the live site
- Exercise 3: Update the site content and confirm the change

Project: Static website on Blob Storage diagram



Knowledge check



1 What must you enable on a storage account before it can serve static website content?

- ☐ Blob versioning
- ☐ Static website hosting
- ☐ Azure CDN integration

2 Where does Azure store the files for a static website hosted in Blob Storage?

- ☐ A container named blob-website
- ☐ The root of the storage account
- ☐ A container named \$web

3 What happens when you upload an updated file to the \$web container?

- ☐ You must redeploy the storage account
- ☐ The updated content is available without redeployment
- ☐ The previous version enters a staging state

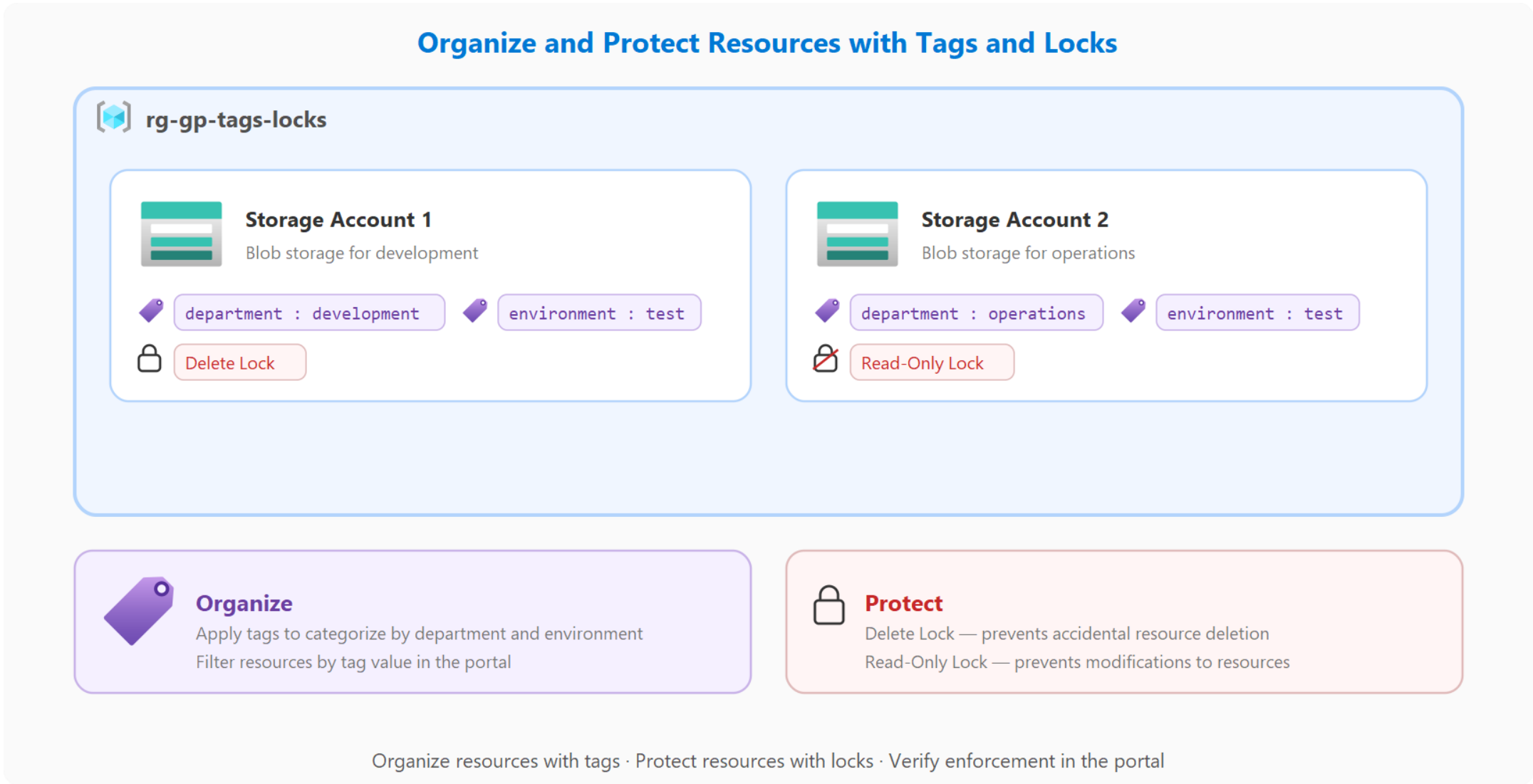
Guided project: Organize and protect resources with tags and locks



Project: Tags and locks for governance objectives

- Scenario: Set up a shared Azure environment with organized and protected resources
- Tags are key-value pairs for cost reporting, ownership tracking, and governance
- Locks prevent accidental modification or deletion, even for users with full permissions
- Exercise 1: Create resources and apply organizational tags
- Exercise 2: Apply resource locks to prevent accidental changes
- Exercise 3: Test lock enforcement and confirm the full lock lifecycle

Project: Tags and locks for governance diagram



Knowledge check



- 1** What is the primary purpose of applying resource tags in Azure?
 - ☐ To restrict who can access a resource
 - ☐ To organize resources for cost tracking and management
 - ☐ To prevent resources from being deleted

- 2** What does a read-only resource lock prevent?
 - ☐ Only deletion of the resource
 - ☐ Modifications and deletion of the resource
 - ☐ All access to the resource including viewing

- 3** What must you do before you can delete a resource that has a delete lock?
 - ☐ Disable the lock using Azure Policy
 - ☐ Contact Azure Support to override the lock
 - ☐ Remove the lock first

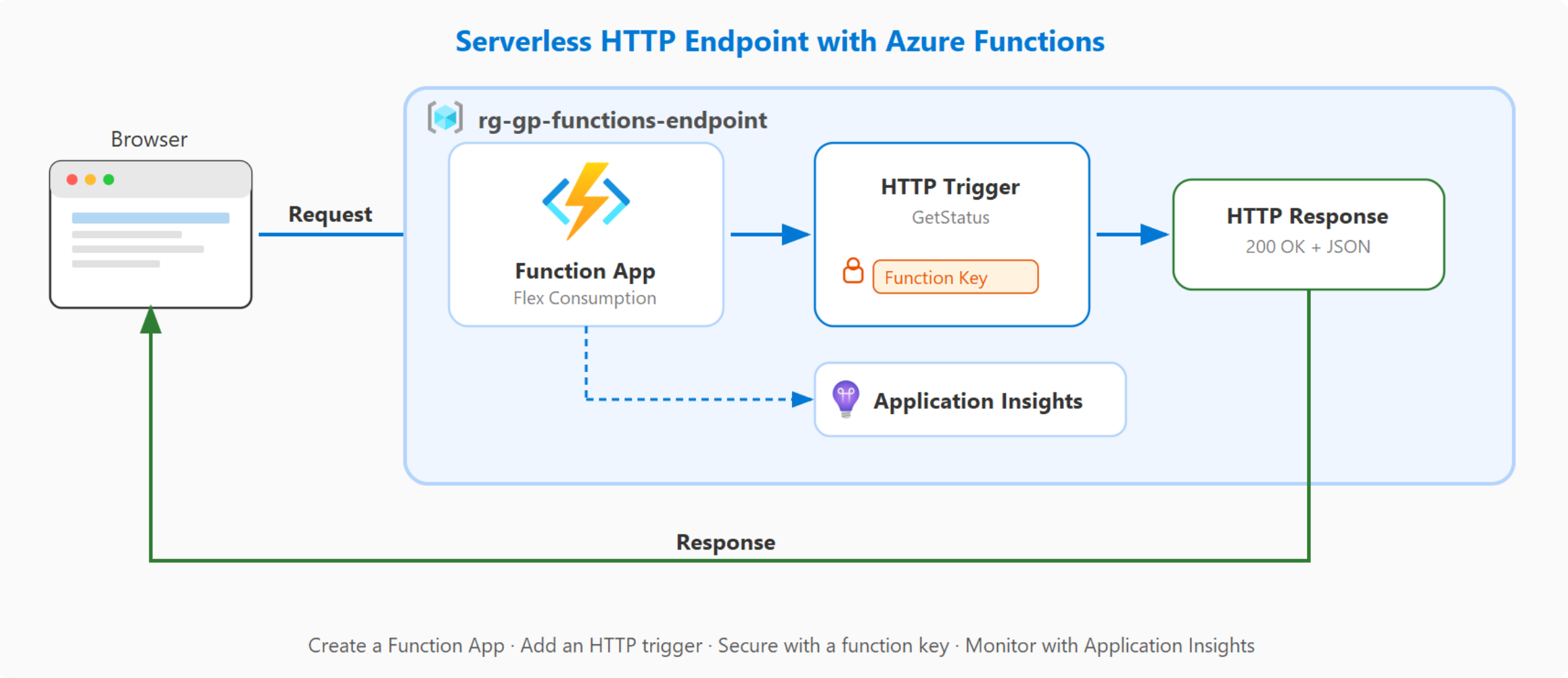
Guided project: Build a simple website endpoint with Azure Functions



Project: Serverless endpoint with Azure Functions objectives

- Scenario: Create a lightweight contact page endpoint without maintaining servers
- Azure Functions is a serverless compute service that runs code in response to events
- An HTTP trigger gives your function its own URL callable from any browser
- Exercise 1: Create a Function App with the Flex Consumption plan
- Exercise 2: Deploy an HTTP-trigger function from Cloud Shell
- Exercise 3: Test the endpoint, add security, and review logs

Project: Serverless endpoint with Azure Functions diagram



Knowledge check



- 1** What type of trigger allows an Azure Function to respond to web requests?
 - ☐ Timer trigger
 - ☐ HTTP trigger
 - ☐ Blob trigger

- 2** What is the main advantage of using Azure Functions for a simple endpoint?
 - ☐ It provides a dedicated virtual machine
 - ☐ You do not need to manage any infrastructure
 - ☐ It automatically creates a custom domain

- 3** What Azure service lets you monitor function invocations and view execution logs?
 - ☐ Azure Monitor metrics
 - ☐ Azure Activity Log
 - ☐ Application Insights

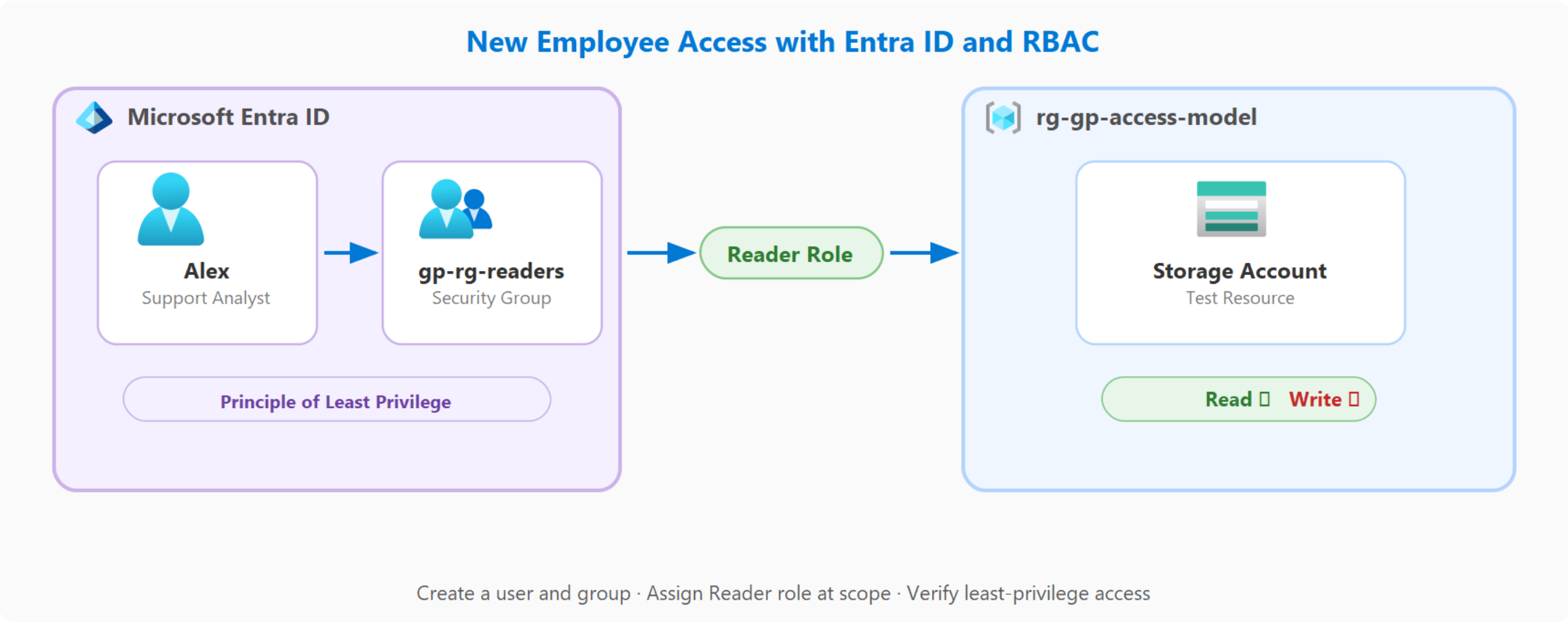
Guided project: Set up new employee access (Entra ID and RBAC)



Project: Identity and RBAC onboarding objectives

- Scenario: A new support analyst needs read-only access to a development resource group
- Microsoft Entra ID manages users and groups; RBAC assigns specific permissions at a scope
- Together they implement the principle of least privilege
- Exercise 1: Create a user account and security group in Entra ID
- Exercise 2: Assign the Reader role to the group at resource group scope
- Exercise 3: Verify read access is granted and write access is denied

Project: Identity and RBAC onboarding diagram



Knowledge check



1 What is the benefit of assigning RBAC roles to a group instead of individual users?

- ☐ Groups automatically inherit subscription permissions
- ☐ It simplifies access management when users join or leave
- ☐ Groups can hold more permissions than individual users

2 What RBAC role provides view-only access to Azure resources?

- ☐ Contributor
- ☐ Owner
- ☐ Reader

3 At what scope level was the RBAC role assigned in this guided project?

- ☐ Subscription
- ☐ Resource group
- ☐ Individual resource

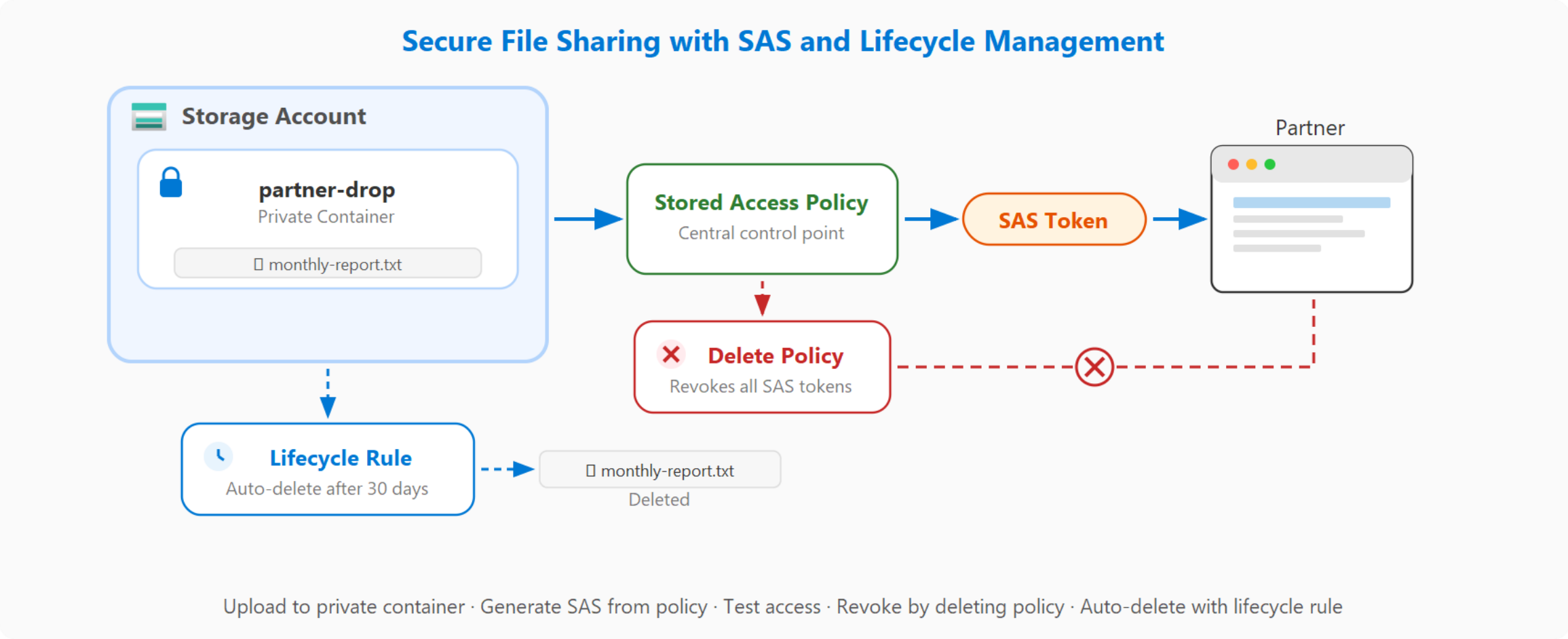
Guided project: Share files securely



Project: Secure file sharing with SAS objectives

- Scenario: Share monthly reports with an external partner for a short review window
- Stored access policies let you centrally manage and revoke SAS tokens instantly
- Exercise 1: Create a storage account, private container, and upload a file
- Exercise 2: Create a stored access policy and generate a SAS link
- Exercise 3: Test SAS access and verify direct access is blocked
- Exercise 4: Revoke access by deleting the stored access policy
- Exercise 5: Configure lifecycle management for automatic cleanup

Project: Secure file sharing with SAS diagram



Knowledge check



1 How can you instantly revoke all SAS tokens generated from a stored access policy?

- ☐ Regenerate the storage account access keys
- ☐ Delete the stored access policy
- ☐ Set the SAS token expiry to a past date

2 What determines the permissions and expiry of a policy-based SAS token?

- ☐ The storage account access tier
- ☐ The stored access policy
- ☐ The container public access level

3 What does a lifecycle management rule do for blob storage?

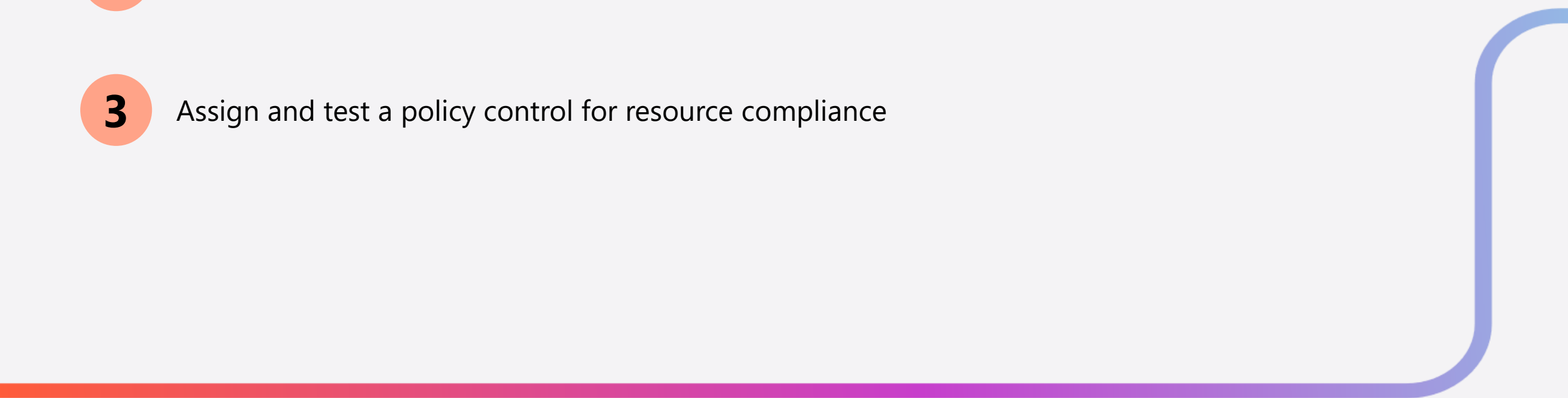
- ☐ Moves blobs between storage accounts
- ☐ Backs up blobs to a secondary region
- ☐ Automatically deletes or moves blobs based on age

Guided project: Set up cost guardrails in Azure



Learning Objectives

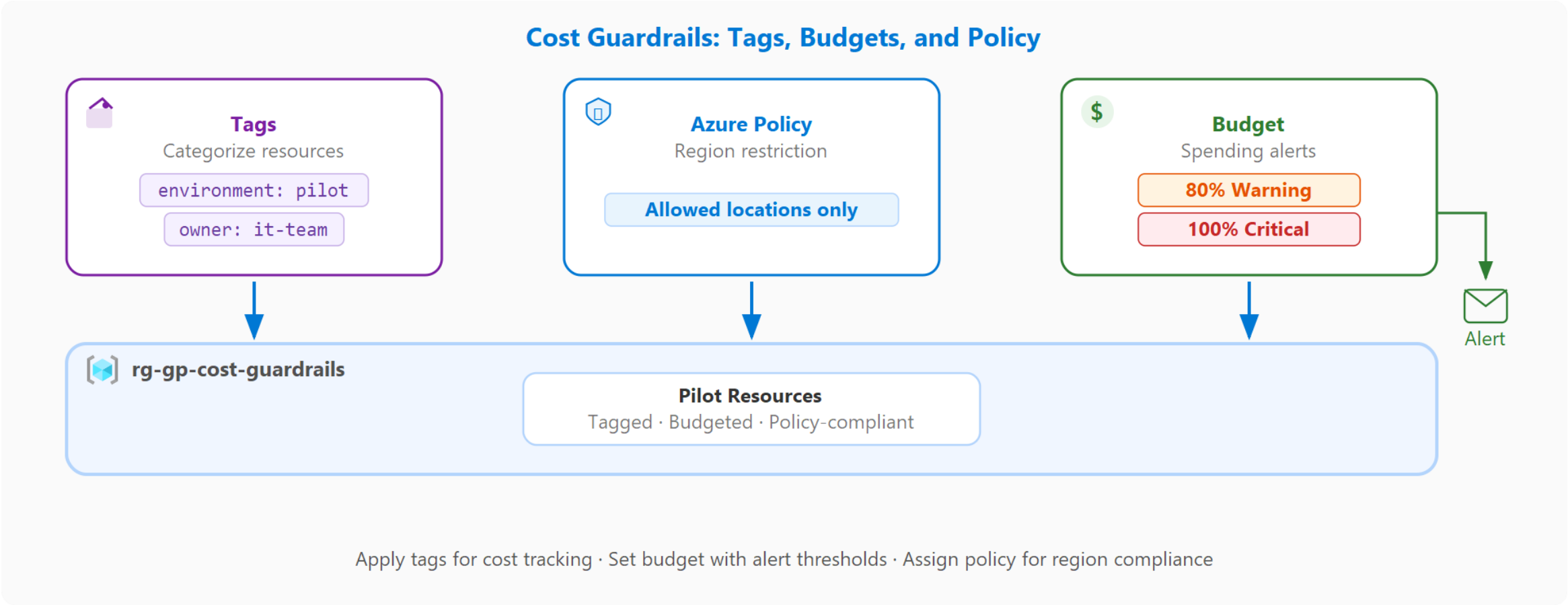
After completing this module, you will be able to:

- 1** Apply consistent resource tags for cost tracking
 - 2** Create budget alerts for early spend visibility
 - 3** Assign and test a policy control for resource compliance
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Project: Cost guardrails with tags, budgets, and policy objectives

- Scenario: A pilot team needs clear cost controls before a broader cloud rollout
- Tags categorize resources for cost reporting; budgets alert when spending reaches thresholds
- Azure Policy enforces organizational rules like restricting deployment regions
- Exercise 1: Apply organizational tags for cost tracking
- Exercise 2: Create a budget with alert thresholds
- Exercise 3: Assign and test an Azure Policy for region compliance

Project: Cost guardrails with tags, budgets, and policy diagram



Knowledge check



1 What happens when a resource deployment violates an Azure Policy with a deny effect?

- ☐ The deployment succeeds but is flagged as noncompliant
- ☐ The deployment is blocked
- ☐ The resource is deployed and then automatically deleted

2 What is the purpose of a budget alert in Azure?

- ☐ To automatically stop all running resources
- ☐ To notify you when spending approaches a defined threshold
- ☐ To enforce a hard spending cap

3 Why should you apply consistent tags to Azure resources?

- ☐ Tags are required before you can deploy resources
- ☐ Tags automatically reduce the cost of resources
- ☐ To track and allocate costs by team, environment, or project

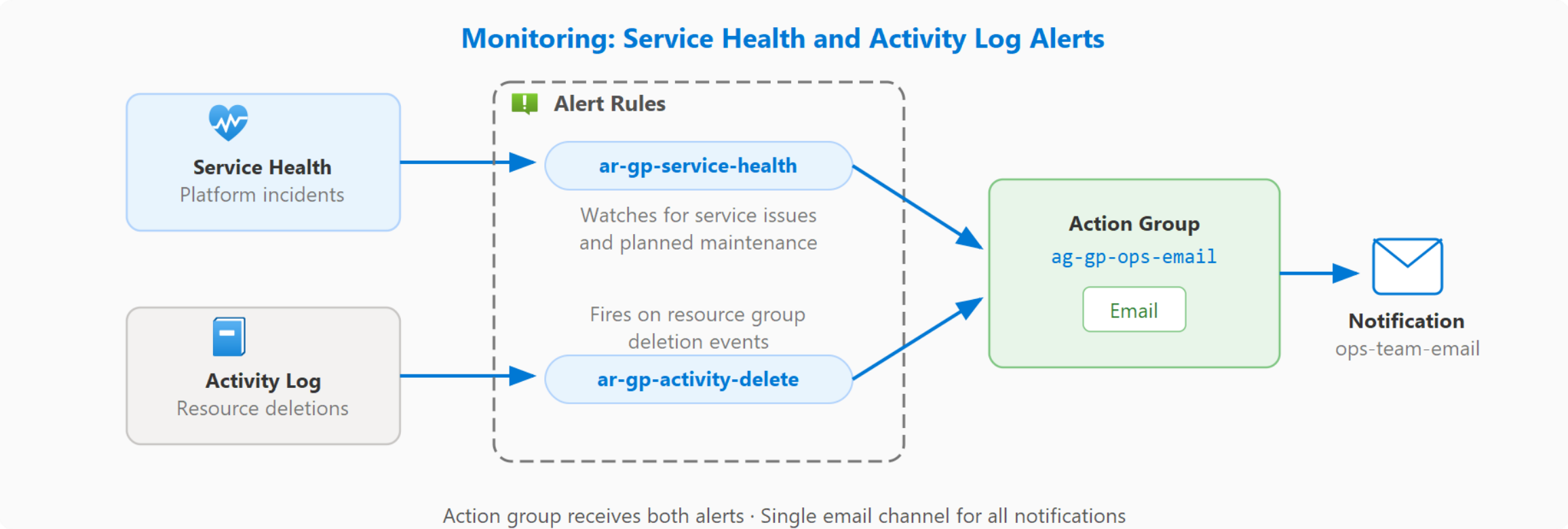
Guided project: Monitor Azure with Service Health and Activity Log alerts



Project: Monitoring with alerts and action groups objectives

- Scenario: Notify the operations team when Azure platform issues occur or resources are deleted
- Action groups define who gets notified and how when an alert fires
- Service Health alerts watch for platform-level incidents and planned maintenance
- Activity Log alerts fire when management operations like resource deletion occur
- Exercise 1: Create an action group and verify email delivery
- Exercise 2: Create a Service Health alert for platform incidents
- Exercise 3: Create an Activity Log alert for resource deletions

Project: Monitoring with alerts and action groups diagram



Knowledge check



- 1** What is the purpose of an action group in Azure Monitor?
 - ☐ To define which metrics are collected
 - ☐ To define who gets notified and how when an alert fires
 - ☐ To group related alert rules into a dashboard

- 2** What type of alert monitors Azure platform incidents and planned maintenance?
 - ☐ Activity Log alert
 - ☐ Metric alert
 - ☐ Service Health alert

- 3** What type of event does an Activity Log alert detect?
 - ☐ Changes in resource performance metrics
 - ☐ Management operations such as resource creation or deletion
 - ☐ User sign-in attempts to the Azure portal

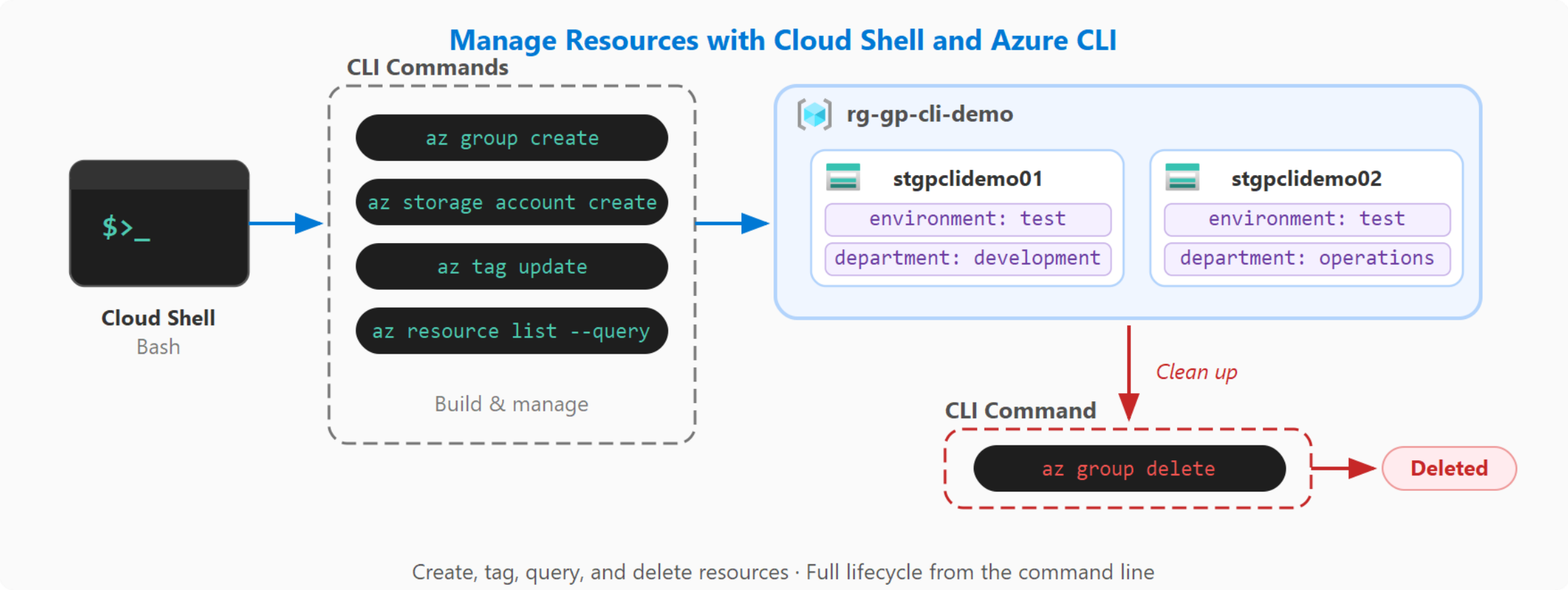
Guided project: Manage Azure resources with Cloud Shell and the Azure CLI



Project: Resource management with Cloud Shell and CLI objectives

- Scenario: Manage resources efficiently from the command line instead of navigating the portal
- Cloud Shell is a browser-based terminal with the Azure CLI preinstalled
- The CLI is useful when performing the same operations across multiple resources quickly
- Exercise 1: Open Cloud Shell and explore the environment
- Exercise 2: Create and list resources with CLI commands
- Exercise 3: Tag, query, and clean up resources from the command line

Project: Resource management with Cloud Shell and CLI diagram



Knowledge check



- 1** What query language does the Azure CLI use to filter and shape JSON output?
 - ☐ SQL
 - ☐ JMESPath
 - ☐ KQL (Kusto Query Language)

- 2** What happens when you delete a resource group using the Azure CLI?
 - ☐ Only the resource group is deleted
 - ☐ All resources inside the resource group are also deleted
 - ☐ Resources are moved to a default resource group

- 3** What shell environment does Cloud Shell use by default for Azure CLI commands?
 - ☐ PowerShell
 - ☐ Command Prompt
 - ☐ Bash

End of presentation



